When Does the Undercut Work?

Pit-Stop Battles in Formula 1, 2023–2026

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Abstract

The undercut, where a car stuck behind a rival pits first to jump ahead on fresh tyres, is a familiar Formula 1 strategy, but we found no public measure of its success rate. Using lap timing from every dry Grand Prix from 2023 to mid-2026, we find 177 green-flag undercut attempts, where the chaser was within 3 seconds, pitted first, and the car ahead responded within five laps. In 2023–25 half the attempts worked (95% CI 41–59%). Each extra second of gap cut the odds of success by 72%; the undercut worked 36% of the time when the car ahead responded the next lap but 72–75% when it waited three laps or more. The commentators' rule of thumb (time gained on fresh tyres before the response must beat the gap) points the right way, but as a yes/no call it is overconfident: on held-out seasons its Brier score (0.331) is worse than always predicting the average (0.252). As a probability, the same rule predicts as well as a full logistic model (0.211 vs 0.218). A typical successful undercut gained 4.0 s on fresher tyres and 1.1 s from the rival's slower stop. Exploratory analyses found that successful undercuts usually held to the flag (worth about 5 points in points fights) and that, before the stop, the gap alone predicted as well as any richer model; car pace, pit crews, tyre hardness and the circuit added nothing. In 2026, 36% of 22 attempts worked, a suggestive but inconclusive drop.

1 Introduction

On lap 9 of the 2024 Las Vegas Grand Prix, Charles Leclerc came in for hard tyres. He had been 0.8 s behind Max Verstappen at the end of the lap before, one lap after Verstappen passed him. Verstappen stayed out until lap 11. By the arithmetic commentators use, Leclerc should have come out comfortably ahead: fresh tyres were worth about 2.0 s a lap at that stage of the race, so two laps on them should have made up 3.9 s, almost five times the gap. At the end of lap 13, with both stops done, he was 3.8 s behind.

Leclerc was trying the *undercut*. In a dry race every car must stop at least once to change

tyres, and tyres get slower as they wear. A stop costs more than 20 seconds in the pit lane, but both cars pay roughly the same price, so in a fight between two cars what matters is who stops first. A car stuck behind a rival has a problem: overtaking on track is hard, and every lap spent in the rival's turbulent air costs time and tyre life. So the chaser pits first and, on new tyres that grip better, laps faster than the rival can on its worn ones. The rival can respond by pitting on the next lap, but every lap it waits, the chaser gains more. If the chaser has made up the gap by the time the rival rejoins from its own stop, it comes out ahead, having passed without overtaking on the track. The chasing driver "must be within striking range," which is "typically within two seconds" (Palmer, 2019).

Television commentary often treats the undercut as a known quantity, worth a set number of seconds at a given circuit, but we found no public estimate of how often it actually works, or of what decides it. Strategy research in motorsport has built full race simulations (Bekker and Lotz, 2009; Heilmeier et al., 2020a), neural networks that make pit-stop decisions (Heilmeier et al., 2020b) and game-theoretic models of when to stop (Aguad and Thraves, 2024). These approaches need lap-time and tyre models that only teams can calibrate well, and they answer "what should a team do?" rather than "what happens when teams try it?" In this journal, Weissbock and Mills (2025) showed how strongly qualifying position predicts the finish, and listed pit and tyre strategy among the open questions for Formula 1. We found no paper in this journal on pit strategy.

Using free lap-by-lap timing data, we find every attempt in the dry Grands Prix from 2023 to round 17 of 2026 and record whether it worked. We ask five questions:

- **RQ1.** How often does the undercut work, and what decides it: the gap before the stop, how quickly the car ahead responds, tyre age, the tyre compound, the time lost in the pit lane, or how fast tyres wear at that circuit?

- **RQ2.** Does a statistical model predict whether an undercut will work better than the rule of thumb that commentators use (fresh-tyre gain per lap, times the number of laps before the car ahead responds, must exceed the gap)?

- **RQ3 (exploratory).** Did the 2026 rules, which brought new cars and new tyres, change the undercut?
- **RQ4 (exploratory).** What kind of team or car closes the gap: a faster car, one that is easier on its tyres, or a quicker pit crew?
- **RQ5 (exploratory).** Can an undercut's success be predicted before the chaser stops, and do factors beyond the gap, such as tyre hardness, track temperature or traffic, help?

In 2023–25 the undercut worked half the time. The commentators' arithmetic is overconfident as a yes/no prediction, but a calibrated version of the same rule predicts as well as a full statistical model. Splitting each attempt's time into what was won at the stops and what was won on track shows that the stops account for about half of the difference between undercuts that work and undercuts that fail. The definitions, models and metrics for RQ1–RQ3 were fixed and tagged in the project's git history before any model was fitted to real data, and RQ3 was planned as exploratory. RQ4 and RQ5 were added afterwards and are exploratory too. So are two shorter analyses: what an undercut is worth at the flag, and whether it works differently at some circuits.

2 Materials and Methods

2.1 Data

We use every Grand Prix from 2023 to the 2026 Singapore Grand Prix (round 17): 22 races in 2023, 24 in each of 2024 and 2025, and 17 in 2026, for 87 in total. Sprint races are excluded because they have no compulsory pit stop. Timing data come from the FastF1 Python package (Schaefer, 2026), which provides each car's lap times, the time it crossed the timing line at the end of each lap, pit-in and pit-out flags, tyre compound and tyre age, and the track status (green, yellow, safety car, virtual safety car or red flag) on every lap.

All data were read from a local cache, so the analysis can be rerun without network access. A race is *wet*, and excluded, if any driver used intermediate or wet tyres or the weather feed recorded rain during the race. This removes 15 races (four in each of 2023–25 and three in 2026), leaving 72 dry races.

2.2 Undercut attempts

We measure the gap between two cars with the times at which they cross the timing line at the end of the same lap. Comparing cars on the same lap number means a car that has been lapped is never treated as the car ahead. A pit stop counts only if the car changes tyres: the lap after its pit-in lap must show a different compound or a lower tyre age. Drive-through and stop-go penalties, which keep the same tyres, are therefore not stops.

Car B (the *chaser*) and car A (the *leader*) form an *undercut attempt* when:

1. on the lap before B's stop (lap $n-1$), A is the car directly ahead of B, and B crosses the line within 3.0 s of A;

2. B makes a real stop on lap n, and neither car was in or out of the pits on lap $n-1$, nor was A on lap n;

3. A makes a real stop on lap m within the next five laps ($n+1 \le m \le n+5$). The *response delay* is $m-n$ laps;

4. every lap from $n-1$ to the check lap (below) is fully green for both cars, with no yellow flag, safety car, virtual safety car or red flag, and neither car uses intermediate or wet tyres or retires before the check lap.

The attempt *works* if B crosses the line ahead of A at the end of lap $m+2$, the lap after A's out-lap, when both stops are complete. If A does not stop within five laps, the case is a *no response* and is counted but not modelled, because A is then on a different strategy rather than responding.

As robustness checks we repeat the analysis with gap windows of 2 s and 5 s, with the outcome checked one lap later (lap $m + 3$), and with the green-flag rule applied only to laps $n - 1$ to $m + 1$.

To check the detection, ten attempts were drawn at random and each one was checked against the two cars' lap times, pit flags, tyre data and line-crossing times (`notes/spotcheck.md` in the repository). All ten were correctly identified and correctly labelled.

2.3 Features

Each attempt is described by the gap on lap $n - 1$; the response delay, grouped as 1, 2, 3 or 4–5 laps; the difference in tyre age between the leader and the chaser on lap $n - 1$ (positive when the leader's tyres are older); and the chaser's compound change (to a harder, the same, or a softer compound). The response delay is the leader's choice, made after it has seen the chaser stop, and it is known only once the leader pits. The RQ1 and RQ2 models therefore describe and predict the outcome *given* the leader's response; RQ5 asks what can be predicted before the chaser stops. Three more features describe the race, and are measured only on the *other* cars in that race, so that no feature uses the attempt's own outcome:

- **Fresh-tyre gain** (s per lap): for each other car's stop within ten laps of lap n, the mean of its two laps before the in-lap minus the mean of its two laps after the out-lap, all green. The race value is the median over those stops.

- **Pit loss** (s): for each other car's green stop, its in-lap plus out-lap time minus twice its median normal lap time nearby. The race value is the median.

- **Tyre wear** (s per lap of tyre age): for each other car's stint of at least six green laps, excluding lap 1 and in- and out-laps, the slope of lap time against tyre age from a

straight-line fit. The race value is the median. Because fuel burns off during a stint, this slope is wear net of the fuel effect.

One 2026 attempt had no other-car stop within ten laps and therefore no fresh-tyre gain; it is dropped from all models, leaving 176 attempts.

2.4 The rule of thumb

Commentators reason about the undercut with simple arithmetic: if fresh tyres are worth g seconds a lap and the leader stays out for d laps, the chaser gains about $g \times d$ seconds and the undercut should work when that is larger than the gap. We define the rule's *margin* as

$$\text{margin} = g \times d - \text{gap},$$

with g the race's fresh-tyre gain from other cars and d the response delay. The rule predicts success when the margin is positive.

2.5 Statistical analysis

RQ1. We estimate success rates overall and by response delay, gap and compound change, with 95% confidence intervals from 2,000 bootstrap resamples of whole races. We then fit a logistic regression of success on the gap, the response delay (categorical), the tyre-age difference, the compound change, the pit loss and the tyre wear, with standard errors clustered by race. Coefficients are reported as odds ratios. This analysis uses 2023–25, the seasons before the 2026 rule change.

RQ2. We compare four ways of predicting whether an attempt works: the *base rate* (the training seasons' success rate for every attempt); the *rule* (a yes/no prediction, margin > 0); the *calibrated rule*, a logistic regression of success on the margin with two parameters, β_0 and β_1; and the full *logistic model* from RQ1. Each is fitted on two of the 2023–25 seasons and tested on the third (leave one season out), then fitted on all of 2023–25 and tested on 2026.

Predictions are scored with accuracy at a 50% threshold and with the Brier score (Brier, 1950), $\frac{1}{N}\sum_i (p_i - y_i)^2$ for predicted probability p_i and outcome $y_i \in \{0, 1\}$, where lower is better. Differences between methods get 95% confidence intervals from 2,000 race-level bootstrap resamples, holding each method's predictions fixed. We also report the calibrated rule's 50/50 margin, $-\hat{\beta}_0/\hat{\beta}_1$, with a race-bootstrap interval, and repeat it with the cold out-lap left out of the fresh-tyre laps (a margin of $g \times (d-1) - \text{gap}$).

RQ3. Because only 22 attempts come from 2026, RQ3 is exploratory. We compare the 2026 success rate with 2023–25 using a race-level bootstrap, add a 2026 indicator to the RQ1 model fitted on all four seasons, and check whether the 2023–25 model's average prediction for 2026 matches what happened.

RQ4. For each attempt we record both cars' teams (merging renamed teams: AlphaTauri, RB and Racing Bulls; Alfa Romeo, Kick Sauber and Audi) and three car traits, each the leader's value minus the chaser's so that a positive number favours the chaser: race pace (the car's median clean lap), its own tyre-wear slope, and its pit-crew time (stop loss as defined below). Each trait is measured relative to the field in every dry race and then averaged, by median, over the team's *other* races in the same season. A trait taken from the attempt's own race would see the outcome: a chaser that wins the undercut then runs in clean air and looks faster. As a check on day-to-day form, a second pace measure uses only the laps of that race before lap $n-1$. Each trait is added to a logistic regression with the gap and the response delay, with standard errors clustered by race. We also test whether adding the chaser's team, or the leader's team, improves a gap-and-delay model (likelihood-ratio test, 9 degrees of freedom).

RQ5. A team deciding whether to stop does not yet know when the leader will respond, so the RQ1 model, which uses the response delay, cannot guide that decision. We therefore compare models that use only what is known before the stop: the gap alone; the RQ1 factors without the delay; and ridge-penalised logistic regressions (Hoerl and Kennard, 1970) with eight pre-stop factors (the RQ1 factors without the delay, plus the fresh-tyre gain, race

progress and whether the chaser is coming off the soft tyre). The ridge penalty is chosen by leaving one season out *within* the training seasons, so the test season is never seen. We also train each model on a wider set of attempts (gaps up to 5 s and local yellow flags allowed: 252 attempts in 2023–25 rather than 154), always testing on standard attempts. For the moment after the leader has responded, we compare the gap and the delay alone with the full RQ1 model and the ridge equivalents. Scoring follows RQ2. Finally, we screen fifteen further factors, adding each on its own to the gap-only and the gap-and-delay models: track temperature; traffic where the chaser or the leader would rejoin (other cars within 2 s of one pit loss behind it); the chaser's closing rate over the last three laps; the trend of the leader's last five lap times; whether the leader is within 1 s of the car ahead; the chaser's position; a street circuit; teammates; a first stop; and five measures of tyre hardness from Pirelli's compound numbers (C1 hardest to C6 softest): the chaser's old and new tyres, the change between them, the leader's tyre and the softest compound at the race. Each race's nomination comes from its Wikipedia article, which cites Pirelli; 16 races were checked against Pirelli or news sources, and all agreed. A factor counts as useful only if it improves the held-out Brier score with a 95% interval that excludes zero and does not make the 2026 score worse, a bar set before scoring. Last, we ask whether the circuit matters. Circuits with fewer than four attempts in 2023–25 are pooled as one group, and the circuit is added to the gap-and-delay model. We test it with a likelihood ratio and with a permutation test that reshuffles which circuit each race was held at (1,000 shuffles), and score it on held-out seasons as above.

Where the time comes from. For each attempt we split the total time the chaser gained on the leader between lap $n-1$ and the check lap (the *swing*) into two parts. The *stop difference* is how much more time the leader's stop cost than the chaser's, each measured as in-lap plus out-lap minus twice the car's normal lap time. The *tyre phase* is the rest of the swing: the laps the chaser spent on fresh tyres while the leader was on old ones. This split is descriptive and is not used in any model.

What an undercut is worth. For each attempt we take the chaser's running position on lap $n-1$ from the timing data and both cars' finishing positions from the official classification; a disqualified car counts as unclassified. Points follow the 25–18–15–12–10–8–6–4–2–1 scale, without the fastest-lap point that 2023–24 also awarded. We report whether the chaser finished ahead of the leader, the places it gained from lap $n-1$ to the flag, and, for attempts in which the leader was in the top ten on lap $n-1$, the points swing within the pair (the chaser's points minus the leader's). We also compare success rates by the chaser's running position (P1–3, P4–6, P7–10 and P11 or lower) and add a top-six indicator to the gap-and-delay model. The position groups were chosen after a first look at the data.

Pre-registration. The attempt definition, features, models and metrics were committed and tagged `prereg` in the repository before any model was fitted to a real attempt. The only earlier look at outcomes was a rough count of attempts and success rates by response delay when the topic was chosen. Deviations, all logged in `notes/decisions.md`, are of two kinds:

- *Changes to pre-registered analyses*, none of which alter the attempt definition, models or metrics: a data-type bug fix; added uncertainty intervals; the time split above, added after the spot-check showed that a stop's time divides between in-lap and out-lap differently with each circuit's pit-lane layout; the no-response group reported as a count only; a rescaled odds ratio for tyre wear; and the compound held at its most common choice in the lookup chart.

- *Later, exploratory analyses*: RQ4 (six tests), RQ5, and the analyses of what an undercut is worth, of running position and of circuits. A first RQ4 pace measure, taken from the attempt's own race, leaked the outcome and showed a strong effect; it was replaced by the season-level measure before anything was reported. RQ5 compares many models and factors on the same seasons, so its best held-out scores are slightly optimistic, and 2026 is the cleaner check.

3 Results

3.1 The attempts

We found 177 undercut attempts in 52 of the 72 dry races (48 in 2023, 59 in 2024, 47 in 2025 and 23 in 2026), and 176 have complete features. In another 314 cases the chaser pitted but the leader did not stop within five laps; these no-response cases are not modelled. In 2023–25 the typical attempt started from a gap of 1.6 s (median), at a race where fresh tyres were worth 1.4 s a lap and a stop cost 22.4 s. The chaser most often moved to a harder compound (92 of 154 attempts).

3.2 RQ1: How often does the undercut work?

In 2023–25, 77 of 154 attempts worked, a success rate of 50% (95% CI 41–59%). When the leader responded on the very next lap (Figure 1, left), the undercut worked 36% of the time (CI 25–49%, $n = 77$). That rose to 50% when the leader waited two laps ($n = 32$) and to 75% ($n = 16$) and 72% ($n = 29$) when it waited three or four to five laps. The gap mattered just as much (right), with 79% of attempts from within 1 s working ($n = 39$), against 52% from 1–2 s ($n = 65$) and 24% from 2–3 s ($n = 50$).

Table 1 shows the logistic regression. Holding the other factors fixed, each extra second of gap multiplied the odds of success by 0.28 (95% CI 0.15–0.51), a 72% drop. Compared with a next-lap response, a response four or five laps later multiplied the odds by 5.4 (CI 1.8–16.5). Tyre age also mattered: each lap that the leader's tyres were older than the chaser's raised the odds by 15% (CI 3–28%). So did tyre wear: at circuits where lap times rose 0.01 s more per lap of tyre age, the odds were 18% higher (CI 5–32%). The pit-lane time loss and the choice of compound made no clear difference.

Figure 2 turns the model into a lookup chart for an average race, read as "if the leader responds after this many laps". With a next-lap response, the undercut is better than even only from about 1.2 s or closer: the model gives 71% at 0.5 s, 57% at 1.0 s, 41% at 1.5 s

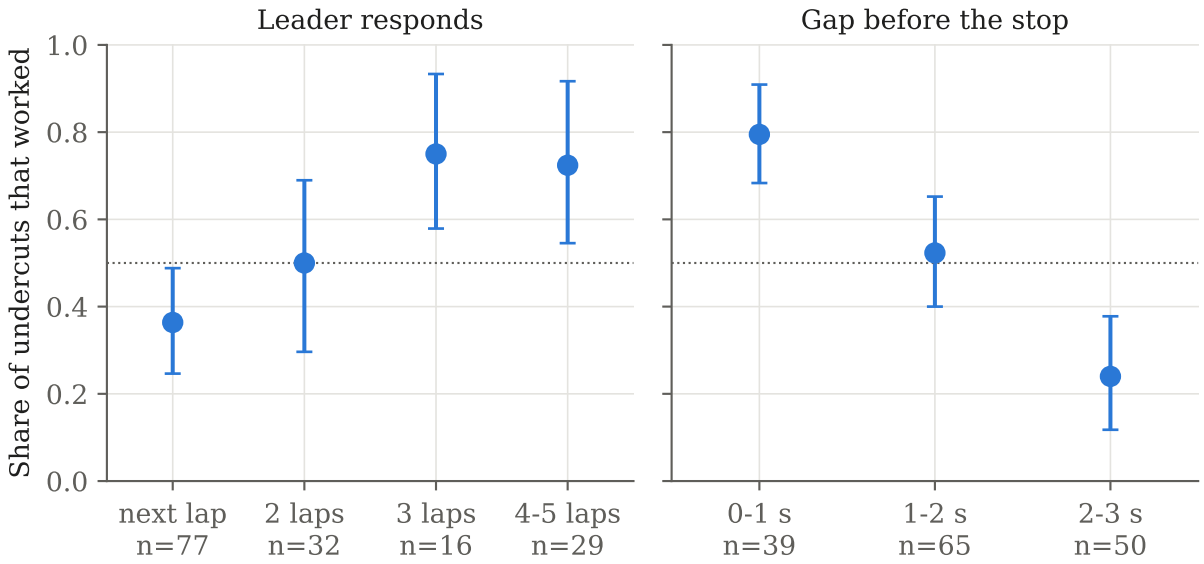


Figure 1: Share of undercut attempts that worked in 2023–25, by how many laps later the leader pitted (left) and by the gap before the chaser's stop (right). Bars are 95% race-bootstrap confidence intervals; n is the number of attempts. The dotted line marks 50%.

and 27% at 2.0 s. If the leader waits three laps or more, the chance at 2.0 s rises to 67–72%. The "striking range" of about two seconds that commentators describe (Palmer, 2019) is therefore a range in which an undercut is possible, but likely only if the leader is slow to respond.

The results hold up under the robustness checks. The odds ratio for each second of gap was 0.31 (CI 0.12–0.79) with a 2 s window, 0.38 (0.29–0.51) with a 5 s window, 0.23 (0.12–0.44) when the outcome was checked a lap later, and 0.26 (0.14–0.48) with the relaxed green-flag rule. The odds ratio for a four-to-five-lap response ranged from 5.4 to 10.4. Success rates moved as expected with the window: 63% within 2 s, 37% within 5 s.

3.3 Where the undercut's time comes from

Figure 3 splits each attempt's swing into the stops and the tyre phase, for the 153 attempts in which both stops' time loss could be measured. Successful undercuts gained 6.0 s on average: 4.2 s in the tyre phase and 1.8 s because the leader's stop cost more than the

Table 1: Logistic regression of undercut success, 2023–25 (154 attempts, 41 races). Odds ratios with 95% confidence intervals from race-clustered standard errors. Reference levels: next-lap response; chaser moved to a harder compound.

Factor	Odds ratio	95% CI	p
Gap before the stop (per s)	0.28	0.15–0.51	< 0.001
Leader responds 2 laps later	1.57	0.62–3.98	0.34
Leader responds 3 laps later	6.97	0.89–54.8	0.07
Leader responds 4–5 laps later	5.45	1.80–16.5	0.003
Tyre-age difference (per lap)	1.15	1.03–1.28	0.01
Same compound (vs harder)	0.70	0.24–2.08	0.52
Softer compound (vs harder)	0.95	0.32–2.84	0.93
Pit loss (per s)	1.06	0.94–1.20	0.37
Tyre wear (per 0.01 s/lap)	1.18	1.05–1.32	0.005

chaser's (medians 4.0 s and 1.1 s). Failed undercuts lost 1.5 s: the tyre phase gained only 0.3 s, and the chaser's own stop cost 1.8 s more than the leader's (medians 0.1 s and −1.1 s). Of the 7.5 s average difference between successes and failures, 3.6 s (about half) came from the stops and 3.9 s from the tyre phase. A few very slow stops pull the stop averages away from the medians: in the 17 attempts where the leader's stop was more than 3 s slower than the chaser's, 15 undercuts worked (88%). Rejoining close behind another car was about equally common in successes and failures (26% and 22%).

3.4 What an undercut is worth (exploratory)

A successful undercut usually held to the flag. After 79% of successes the chaser was still ahead of the leader at the flag (CI 70–89%), against 17% after a failure (CI 9–27%). A failed attempt cost the chaser a median of one place between the lap before its stop and the finish (mean 2.2 places), while a successful one cost none (mean 0.1). In the 107 attempts in which the leader was running in the top ten, the points swing within the pair was +1.5 after a success and −3.7 after a failure, a difference of 5.2 points (CI 3.3–7.1), although cars whose undercuts worked may simply have been faster.

Undercuts worked less often near the front, 40% of the time for a chaser running in the

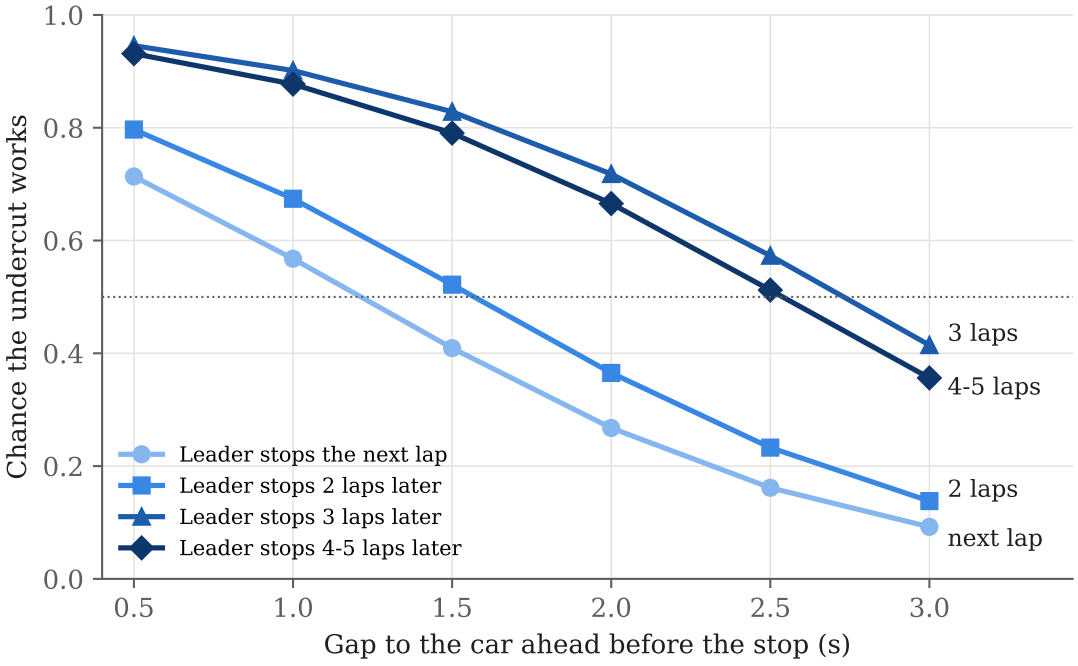


Figure 2: Predicted chance that an undercut works, by the gap before the chaser's stop and how many laps later the leader pits. From the logistic regression in Table 1, with tyre-age difference, pit loss and tyre wear at their 2023–25 medians and the chaser moving to a harder compound, the most common choice.

top three ($n = 20$) and 29% in fourth to sixth ($n = 31$), against 57% in seventh to tenth ($n = 42$) and 59% further back ($n = 61$). Front-runners tried from larger gaps (median 1.9 s against 1.4 s), but with the gap and the response delay held fixed, a chaser in the top six still had lower odds of success (odds ratio 0.41, CI 0.16–1.04, $p = 0.06$; groups chosen after a first look at the data).

3.5 RQ2: Does a model beat the rule of thumb?

Read as a yes/no prediction, the rule of thumb is right 67% of the time in 2023–25. When it said the undercut would work, it did 64% of the time ($n = 92$); when it said it would fail, 29% still worked ($n = 62$). Leclerc's stop in Las Vegas was one of the 33 attempts the rule called a success that failed: his new hard tyres lost 2.8 s to Verstappen over the tyre phase, so the gain the rule counted on never appeared. In the calibrated rule, the margin is clearly

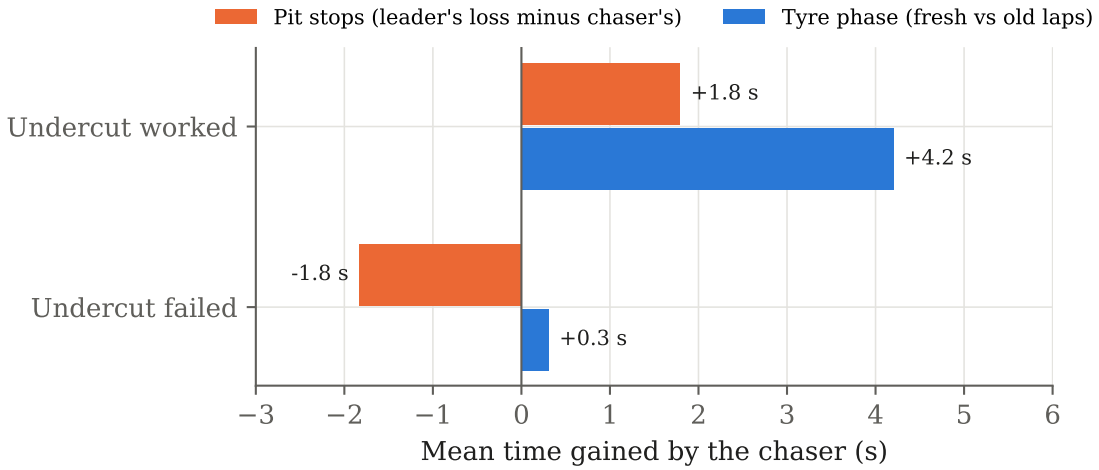


Figure 3: Mean time the chaser gained on the leader, split into the pit stops (the leader's stop time loss minus the chaser's) and the tyre phase (everything else between the lap before the chaser's stop and the check lap). 2023–25, the 76 successful and 77 failed attempts in which both stops' time loss could be measured.

linked to success (each extra second of margin multiplies the odds by 1.51, CI 1.23–1.86), but its 50/50 point is uncertain. With the fresh-tyre laps counted as in the rule above, an attempt is a coin flip at a margin of 1.0 s (95% CI −0.1 to 1.9 s); leaving the cold out-lap out of the count moves this to −0.3 s (CI −1.3 to 0.6 s).

Figure 4 and Table 2 compare the four methods on seasons they were not fitted on. The yes/no rule is about as accurate as the fitted methods (67%) but has the worst Brier score (0.331), worse even than predicting the base rate for everyone (0.252; difference +0.079, CI +0.009 to +0.145), because it is confidently wrong a third of the time. The calibrated rule had a Brier score of 0.211, better than the base rate (difference −0.041, CI −0.073 to −0.012). The full logistic model, with ten fitted parameters instead of two, did no better (0.218; difference from the calibrated rule +0.007, CI −0.030 to +0.046). It was the best method when 2023 was held out but scored worse than the base rate when 2025 was held out (0.287 vs 0.253).

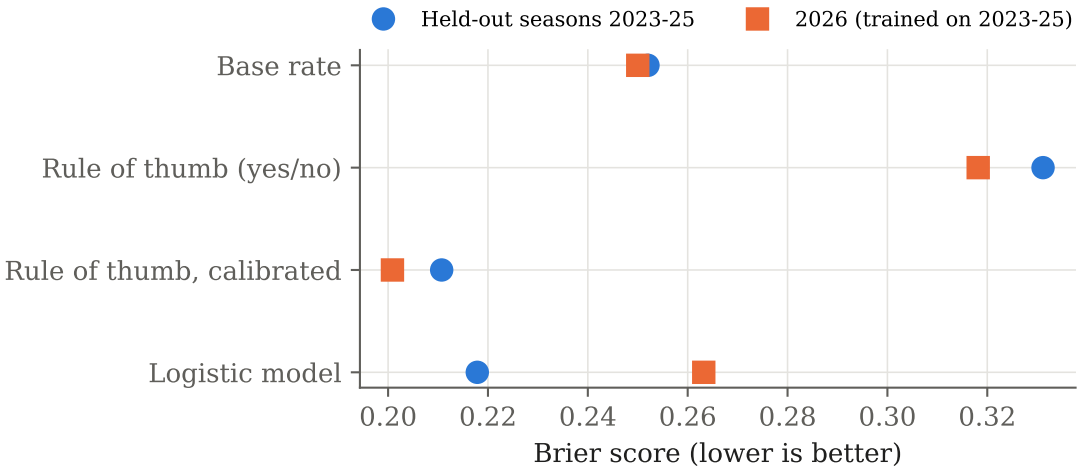


Figure 4: Brier score of four predictions of whether an undercut works (lower is better). Circles: each 2023–25 season predicted by a model fitted on the other two (154 attempts). Squares: 2026 predicted by a model fitted on 2023–25 (22 attempts).

Table 2: Out-of-sample performance. Leave one season out over 2023–25 (pooled, 154 attempts) and a 2023–25 model applied to 2026 (22 attempts).

	2023–25, season held out		2026	
Method	Brier	Accuracy	Brier	Accuracy
Base rate	0.252	0.47	0.250	0.36
Rule of thumb (yes/no)	0.331	0.67	0.318	0.68
Rule of thumb, calibrated	0.211	0.68	0.201	0.73
Logistic model	0.218	0.67	0.263	0.64

3.6 RQ3 (exploratory): Did 2026 change the undercut?

In 2026, 8 of 22 attempts (in 10 races) worked, or 36%, against 50% in 2023–25. The difference, −14 percentage points, has a 95% CI of −36 to +3 points. With the other factors held fixed, a 2026 indicator gave an odds ratio of 0.34 (CI 0.11–1.00, $p = 0.049$), and the 2023–25 model predicted an average success rate of 56% for the 2026 attempts, against the 36% that happened. Leaders also responded faster: 14 of the 22 responses (64%) came on the next lap, against 77 of 154 (50%) before, and of those next-lap cases only 3 of 14 worked (21%).

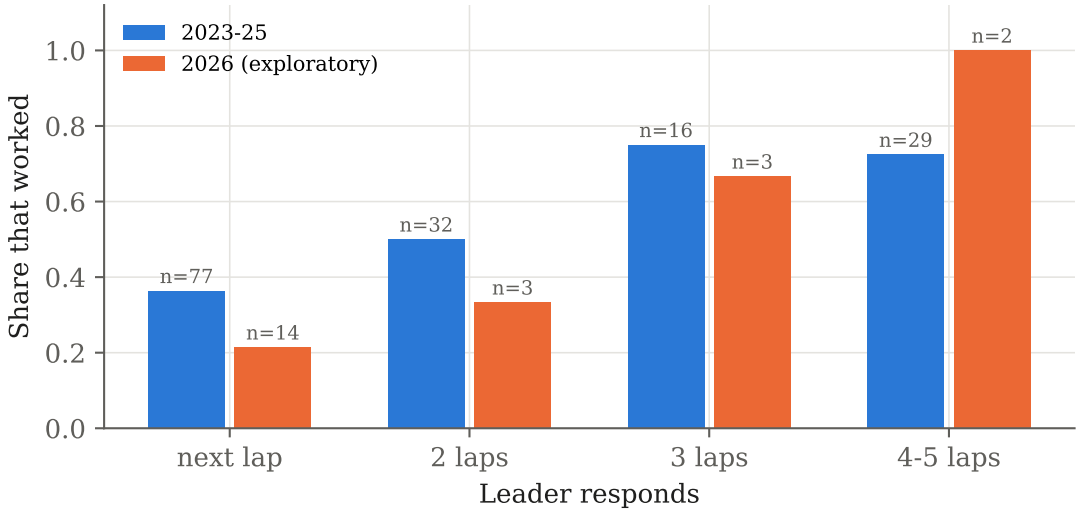


Figure 5: Share of undercut attempts that worked, by response delay, in 2023–25 and in 2026 (exploratory; n above each bar).

3.7 RQ4 (exploratory): What kind of team or car closes the gap?

Faster cars did better at first sight: when the chaser's team was the faster one over the season's other races, 58% of its undercuts worked ($n = 69$), against 44% when it was not ($n = 85$). But faster cars also tended to start closer (correlation -0.17 between the pace advantage and the gap), and once the gap and the response delay were in the model, pace added nothing: an odds ratio of 1.00 per 0.1 s a lap (CI 0.89–1.13). Pace on the day, from the laps before the attempt, told the same story (59% vs 42% raw; odds ratio 1.01 per 0.1 s a lap, CI 0.86–1.19, $n = 146$). Neither a car's tyre wear (odds ratio 1.00 per 0.01 s a lap, CI 0.66–1.50) nor its team's pit-crew record (1.65 per second, CI 0.70–3.86) was clearly linked to success. The crew record did not even predict how the attempt's own stops went (correlation 0.07 with the stop difference), so the slow stops that decide some undercuts look like one-off events rather than a team trait.

Which team was being undercut was linked to success. Adding the leader's team to a gap-and-delay model improved it ($p = 0.017$); adding the chaser's team did so only weakly ($p = 0.06$). Figure 6 shows the raw rates. Undercuts against Aston Martin worked 4 times

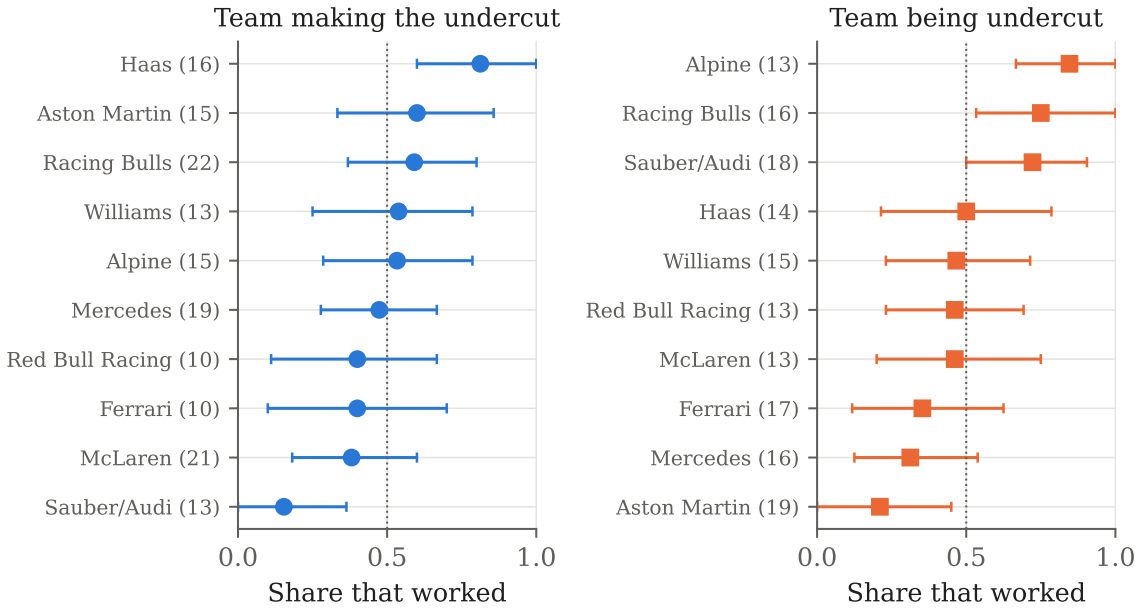


Figure 6: Share of undercut attempts that worked, by the team making the undercut (left) and the team being undercut (right), 2023–25 (exploratory). Number of attempts in brackets; bars are 95% race-bootstrap intervals. Each team's cars changed over the three seasons.

in 19, against Mercedes 5 in 16 and against Ferrari 6 in 17, but against Alpine 11 in 13, Racing Bulls 12 in 16 and Sauber/Audi 13 in 18. As the chaser, Haas succeeded 13 times in 16 and Sauber/Audi 2 times in 13. Every team has fewer than 25 attempts, so the intervals are wide.

3.8 RQ5 (exploratory): Can an undercut be predicted before the stop?

Before the stop, the gap alone was the best predictor (Table 3): a Brier score of 0.209 on held-out seasons, against 0.252 for the base rate (difference −0.043, CI −0.075 to −0.012). Its probabilities matched what happened: attempts it gave a 70% chance or more worked 83% of the time, and those it gave under 30% worked 20% of the time. Every richer pre-stop model did worse. Adding the tyre-age difference, compound choice, pit loss and tyre wear raised the score to 0.234 (difference from the gap alone +0.025, CI −0.001 to +0.054), and

shrinking eight factors with ridge regression did not rescue it. In 2026, when fewer undercuts worked, no pre-stop model clearly beat the base rate. After the response, the gap and the delay together (0.200) did at least as well as the full RQ1 model (0.218; difference −0.018, CI −0.050 to +0.012) and every other method in this paper. They also held up in 2026 (0.212, against 0.250). Training on the wider set of attempts changed little.

None of the fifteen extra factors met the bar. Two were associated with success in 2023–25 but did not improve predictions: a chaser that had been closing in over the last three laps (odds ratio 3.1 per second a lap, CI 1.1–8.9, $p = 0.03$) and, less clearly, a street circuit (0.47, CI 0.19–1.14, $p = 0.10$). With fifteen factors tested, one p-value below 0.05 is not unusual by chance. Tyre hardness showed no association at all: each step softer for the chaser's new tyre gave an odds ratio of 0.82 (CI 0.56–1.19), and for its old tyre 1.00 (CI 0.63–1.60).

Adding the circuit to the gap-and-delay model improved the fit in 2023–25 (likelihood ratio $p = 0.008$; permutation $p = 0.015$). The undercut did work more often at some circuits than the gap and delay explain: 6 times in 6 at Imola, 5 in 5 in Spain and 14 in 21 in Bahrain, against 1 in 8 in Qatar, 2 in 8 in Abu Dhabi and 5 in 15 in Austria. But a circuit's record did not carry over to a held-out season: the Brier score got worse, from 0.200 to 0.232 (difference +0.033, CI −0.011 to +0.088). Across the 22 circuits, the success rate followed the typical gap of the attempts there (Spearman $\rho = -0.54$, $p = 0.009$) but not the circuit's tyre wear, pit loss or fresh-tyre gain (ρ from 0.13 to 0.21, all $p > 0.3$).

4 Discussion

The yes/no rule is overconfident mainly because it assumes that every lap before the leader responds earns the full fresh-tyre advantage, taken as a median over other cars rather than what this chaser gets on this compound. The first of those laps is the least reliable, since the chaser starts it on cold tyres while the leader can push on its in-lap, and we suspect that a good share of undercuts are decided there. Our data cannot settle whether that lap should

Table 3: Predicting whether an undercut works (exploratory). Brier scores (lower is better) for seasons held out in turn (2023–25, 154 attempts) and for 2026 with a model fitted on 2023–25 (22 attempts). "Wider" models are trained on 252 attempts with gaps up to 5 s and yellow flags allowed, and tested on standard attempts.

Model	2023–25, season held out	2026
Base rate	0.252	0.250
Before the stop		
Gap alone	0.209	0.253
Gap alone, wider	0.211	0.245
RQ1 factors without the delay	0.234	0.296
Ridge, eight factors	0.238	0.276
Ridge, eight factors, wider	0.226	0.279
After the leader responds		
Gap and delay	0.200	0.212
Gap and delay, wider	0.200	0.200
Full RQ1 model	0.218	0.263
Ridge, nine factors	0.227	0.250
Ridge, nine factors, wider	0.201	0.249

count at all. With it, the rule's 50/50 point is a margin of 1.0 s; without it, the point moves to -0.3 s.

With two fitted numbers, the calibrated rule predicted as well as the full model with ten (Brier scores 0.211 and 0.218; Table 2), and the gap and the response delay alone did at least as well as either (0.200; Table 3). Since there are only 154 attempts and 77 successes, every extra coefficient carries noise, and in a new season that noise can cost more than the factor adds. The full model was the best method with 2023 held out but worse than the base rate with 2025 held out. Broadcasters could quote a probability instead of calling the outcome. Teams have more to go on (tyre temperatures, degradation models, the rival's radio and their own plans), and any of these might beat the gap alone.

Of the two numbers that matter, the chaser controls only the gap (Figure 2). From within a second, 79% of undercuts worked, and the model gives better-than-even odds up to about 1.2 s even when the leader responds on the next lap. At 2 s the odds are better than even only if the leader waits three laps or more, so a chaser at the edge of the "striking

range" is relying on a slow response. If leaders waited only when they were safe, undercuts against a slow response should rarely have worked, but 72–75% of them did. Some leaders may be forced to wait by another car they are racing, a teammate already in the pit lane, traffic where they would rejoin or a plan for a longer stint, and others may simply misjudge the threat. Because the leader picks its delay after seeing the chaser stop, we cannot say that a late response causes a failed defence. Game-theoretic models of pit stops (Aguad and Thraves, 2024) treat the undercut as a contest between two drivers' decisions, and our results are consistent with that.

About half of the difference between successful and failed undercuts came from the stops themselves (3.6 of 7.5 s; Figure 3). When the leader's stop was more than 3 s slower, 15 of 17 undercuts worked. A team's pit-crew record did not predict its stops in a given attempt (correlation 0.07), so slow stops look like one-off events. An undercut that failed on a slow wheel change may therefore have been the right call when it was made, and should be judged that way. Race simulations should also treat stop time as random, as Monte Carlo simulations do (Heilmeier et al., 2020a), since fixed stop times would miss an effect of this size.

Faster cars won more undercuts (58% against 44%), but they also started closer, and once the gap and the response delay were in the model, pace was no longer associated with success (odds ratio 1.00 per 0.1 s a lap). The team effect that did show up was on the defending side (Figure 6), though with fewer than 25 attempts per team and six tests added after the main analysis it is weak evidence. Undercuts against Aston Martin, Mercedes and Ferrari rarely worked, and those against Alpine, Racing Bulls and Sauber/Audi usually did. Some teams may be better at parts of a defence that timing data cannot see, such as warming their own new tyres on the out-lap. Front-runners also did worse as chasers. With the gap and the response delay held fixed, a chaser in the top six had lower odds of success (odds ratio 0.41, $p = 0.06$, in groups chosen after seeing the data). The leading teams may watch each rival more closely and respond in ways the delay does not capture.

Circuits mattered mainly through the gap, with undercuts working more often where attempts started closer ($\rho = -0.54$) and no more often where tyre wear or the fresh-tyre gain was larger. A circuit's record also did not improve held-out predictions, which is unsurprising with fewer than ten attempts at most circuits.

In 2026 fewer undercuts worked (36% against 50%; Figure 5) and leaders responded faster, with 14 of the 22 responses on the next lap; only 3 of those undercuts worked. Even with the delay and the other factors held fixed, 2026 attempts had lower odds (odds ratio 0.34), and the 2023–25 model expected 56% success against the 36% observed. The new tyres may take longer to warm up, which would hurt the chaser's out-lap, and teams may also have learned to respond sooner. The calibrated rule and the gap-and-delay model still beat the base rate in 2026 (Brier scores 0.201 and 0.212, against 0.250), so the arithmetic seems to have survived the new rules. We cannot yet say the odds have changed. The interval for the change in success rate runs from -36 to $+3$ points, which includes no change, and the 2026 indicator reaches only $p = 0.049$ in a test we planned as exploratory.

4.1 Limitations

The sample is modest: 154 attempts in 2023–25 and 22 in 2026, so several estimates, especially for three-lap responses and for 2026, have wide intervals. We use only dry races and attempts in which every lap of the stop window was fully green; undercuts under safety cars, which follow different logic, are excluded, and so are attempts with a local yellow flag on any lap of the window. The response delay is chosen by the leader after the chaser stops, so its odds ratios describe association, not the effect of an assigned delay. The definition requires the leader to respond within five laps. Cases where the leader stayed out longer are a different strategic situation (often a different number of stops) and are not modelled. The race features are medians over other cars, so they describe the race, not the exact tyres of the two cars involved, and tyre wear is measured net of fuel burn. The fresh-tyre gain behind the rule of thumb is one reasonable estimator among several, and the rule's break-even point

moves by more than a second depending on whether the out-lap is counted; a team with tyre-model data would estimate it better. The analysis uses only races already in the local data cache (to round 17 of 2026), because the FastF1 timing archive began requiring authentication in October 2026. In RQ4, team results pool three seasons in which each team's car changed, and a crew's stop loss measured from lap times includes the driver's in-lap and out-lap, not only the time stationary. In RQ5, many models and factors were compared on the same seasons, so the best held-out scores are slightly optimistic. The compound numbers come from Wikipedia (checked for 16 races), and what each C-number means shifted between seasons as Pirelli changed its range (a C6 in 2025, new tyres in 2026). The points swing leaves out the fastest-lap point of 2023–24 and counts the two disqualified drivers in our attempts as unclassified, and most circuits have fewer than ten attempts. Finally, the analysis describes what happened when teams chose to undercut; it cannot say what would have happened had they not.

4.2 Reproducibility

All code, the pre-registration tag, the decision log and every result file are at `https://github.com/0909yejun0909-commits/undercut`. From a FastF1 cache of the 2023–2026 races, the pipeline is: `build_laps`, `build_attempts`, `build_teams`, `build_weather`, `rq1`, `rq2`, `rq3`, `rq4`, `rq5`, `rq5b`, `rq5c`, `stakes`, `circuits` and `figures`, each run as `uv run python -m analysis.<step>`. The compound table (`reference/compounds.csv`) was built with `build_compounds`, which needs an internet connection, and is included. A command-line tool gives the chance for a given gap: `uv run python -m analysis.predict --gap 1.5`. The 80 unit tests run with `uv run pytest`.

5 Conclusion

Half of Formula 1's green-flag undercut attempts in 2023–25 worked, and two numbers did most of the deciding: the gap before the stop and how many laps the car ahead waited before responding. The commentators' arithmetic for combining them points the right way, but as a yes/no call it is overconfident. Turned into a probability, the same rule did as well as a full statistical model, and before the stop the gap alone was as good a guide as any model we could build from public timing data. Under the 2026 rules, leaders have responded faster and fewer undercuts have worked. With 22 attempts we cannot yet tell whether the undercut has become harder or the 2026 drop is chance.

Acknowledgments

The author thanks the developers of FastF1 for free access to Formula 1 timing data. AI assistance (Anthropic's Claude) was used to write analysis code, to collect the tyre-compound table, and to draft and edit the text, including the Discussion. The research question, the choice of analyses and all final decisions are the author's.

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